

Lucius Wilbert Tjoa

AI Engineer seeking a full-time, permanent role

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PERSONAL SUMMARY

AI Engineer | Agentic AI, Multimodal AI, RAG, AWS

AI Engineer with 2+ years of experience architecting and building end-to-end intelligent systems across agentic AI, multimodal AI, RAG, and business workflow automation. Experienced in developing LLM-powered applications, integrating NLP and computer vision pipelines, and deploying scalable AWS-based solutions using serverless and cloud-native architectures. Many AI hackathon winner recognized for rapid prototyping and translating AI capabilities into practical business impact.

EDUCATION

Asia Pacific University of Technology & Innovation (APU)

Kuala Lumpur, Malaysia

Bachelor of Computer Science (Hons) (Artificial Intelligence)

Cumulative GPA: 3.82

- Vice Chancellor's List for Excellent Academic Achievement, Academic Years 2023/2024 and 2024/2025

WORK EXPERIENCE

CTI Research & Development (AI Automation Engineer)

Dec 2024 - Present

- Automated business processes by integrating AWS Bedrock with Lambda to execute AI-driven tasks.
- Built serverless workflows using S3 for file storage and DynamoDB for data logging and organization.
- Deployed end-to-end AWS automation solutions to reduce manual operations and improve processing speed.

AI Engineer at CelcomDigi

June - October 2025

- Built and deployed RAG pipelines for LLM-powered knowledge retrieval across enterprise use cases.
- Developed AI automation solutions to improve operational efficiency in real business workflows.
- Integrated LangChain and LiteLLM to support modular orchestration and standardized multi-provider LLM usage.
- Utilized AWS services for scalable deployment of AI applications in cloud environments.

Technical Assistant

November 2023 - Dec 2024

- Fine-tuning and optimizing hardware components to meet performance requirements, ensuring seamless system functionality, efficiency, and reliability through precise configuration and troubleshooting

AWARDS

Great Malaysia AI Hackathon 2025 | Top 20 / 1,000+ participants

Oct 2025

- Developed an AWS-powered AI solution to localize global marketing content for the Malaysian market.

KitaHack Google Developer Group on Campus Hackathon | Top 10 / 600+ participants

Apr 2025

- Led a team of three to build iPocket, an AI-powered expense mobile application.

Joget NextGenHack Hackathon | 1st Place

Nov 2024

- Built an AI-based financial application with a team of four to help companies manage finance-related data.

DevMatch Hackathon | 1st Place

Aug 2024

- Built an AI-based voting system with a team of four.

PROJECTS

AI-Powered Cloud Drive with Multimodal Search & Chatbot | Final Year Project

March 2026 - Ongoing

- Built and deployed Multimodal Retrieval-Augmented Generation (RAG) pipelines for Large Language Models (LLMs) to enhance knowledge-based querying.

- Built default and advanced retrieval modes, enabling fast semantic search for everyday use and deeper context-aware retrieval using vector search, query expansion, reranking, and knowledge graph enrichment.
- Designed multimodal ingestion pipelines that process uploaded content into embeddings for search, while converting document text into a Neo4j knowledge graph to support richer relationship-aware answers.
- Implemented the system on AWS using EC2, Cognito, API Gateway, Lambda, S3, and DynamoDB, with Amazon Bedrock, Qdrant, and Neo4j powering the AI retrieval and chatbot workflows.

Customer Service Agent

May 2026 - Ongoing

- Designing an agentic workflow layer that determines whether to answer directly, create tickets, escalate cases, or initiate service actions such as refund or replacement requests.
- Building an AWS Bedrock-based customer service system that retrieves customer history, prior issues, and payment status to support AI-assisted customer interactions and support workflows.
- Developing tool-based backend actions for customer profile lookup, previous ticket retrieval, payment-status checks, ticket creation, escalation routing, and support case handling.
- Integrating Nova Sonic as a voice interface on top of the backend workflow to support both chat-based and voice-enabled customer service experiences.

End-to-End Store Sales Forecasting & MLOps Pipeline | [Github Link](#)

December 2025 - February 2026

- Developed a time-series forecasting system for Corporación Favorita grocery sales using an ensemble of CatBoost, XGBoost, and LightGBM.
- Engineered features to handle high-cardinality retail data, including cyclic time encoding (sin/cos), holiday mapping, and oil price correlation.
- Orchestrated experiment tracking and model versioning using MLflow on AWS EC2, utilizing Optuna for automated hyperparameter tuning.
- Architected a full-stack deployment on AWS (S3, EC2) with a Streamlit dashboard, enabling real-time "What-If" scenario simulations for promotional and economic shifts.

SKILLS

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- **Language:** English, Malay, Indonesian
 - **Programming Language:** Python, SQL
 - **AI/ML:** Deep Learning, NLP, Computer Vision, PyTorch
 - **LLM & Agentic AI:** RAG, LangChain, Amazon Bedrock, MCP
 - **Cloud & Database:** AWS, MySQL, DynamoDB, Neo4j, Qdrant